Venkata Lakshmi Parimala Pasupuleti

AI Developer · Generative AI · LLMs · RAG Pipelines · Prompt Engineering · Python · REST APIs · AWS

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PROFESSIONAL SUMMARY

AI Developer with 2 years of hands-on Generative AI development — building production LLM applications, RAG pipelines, and enterprise automation solutions in Python. Designed and shipped **Leasel**: a multi-agent Generative AI platform (8-agent LangGraph, RAG, OCR, intent classification, summarization, HITL) achieving 93.6% precision across 1,240 documents — deployed against real enterprise compliance workflows with output validation and data governance enforced from day one. Deep experience integrating LLMs with enterprise systems via REST APIs, building CI/CD-gated evaluation pipelines, and shipping automation solutions on AWS. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Generative AI & LLMs | RAG Architectures | Prompt Engineering | Intent Classification | Python Automation | REST API Integration | Multi-Agent Orchestration | CI/CD & LLM Evaluation | AWS & Cloud Platforms | Data Governance & Privacy | SQL & Relational Databases | Enterprise AI Integration

EXPERIENCE

George Mason University — Research Associate

Aug 2024 – Aug 2025

AI/ML Research · Python · Statistical Modeling · Automation

- Built multi-source prediction models (Python, R) achieving $R^2 = 0.977$ across 190+ countries on WHO and UK Road Safety datasets — full pipeline from data collection and transformation to validation and stakeholder reporting; applied Isolation Forest anomaly detection to automate data integrity checks across 3 research pipelines.
- Designed and delivered AI-powered analytics dashboards (Tableau, ggplot2, Quarto) translating complex multi-country datasets into decision-ready insights — improved research comprehension by 40% across two faculty cohorts.

George Mason University — Graduate Teaching Assistant

Aug 2025 – May 2026 · Part-time

Python Programming · Database Design · SQL Instruction

- Mentored **600+ students** in Python programming (data structures, OOP, automation scripting) and database design (SQL, ER modeling, normalization) — developed walkthroughs and exercises; provided one-on-one debugging support.

Entrata (PropTech / Real Estate SaaS)

Jan 2017 – Aug 2024

Software Engineer — Python · REST APIs · Data Pipelines · AWS · Automation

- **Shipped Entrata's first production AI pipeline** across 1.6M real financial records (Python, PyTorch, OpenAI API) — automated complex data processing workflows with output validation, quality gates, and compliance controls enforced from day one; zero violations at launch.
- Built and consumed **REST APIs** across enterprise integrations (Salesforce CRM sync, AWS S3 data pipelines, RabbitMQ messaging, third-party compliance systems); developed and optimized **200+ SQL queries** (PostgreSQL) for 30% performance gain through execution plan tuning and indexing.
- **Automated enterprise workflows** via Python/Bash pipelines eliminating 80% of manual operational workload; achieved 100% PHI compliance on AWS S3 migration (Libsodium-encrypted, zero security incidents); enforced data governance, access controls, and audit logging across all PII/PHI data.
- Deployed and maintained enterprise software on **AWS (S3, EC2, RDS, Glue)** with CI/CD pipelines (GitHub Actions); contributed to Docker-based microservices architecture redesign (50% scalability improvement); participated in 120+ Agile sprints across cross-functional stakeholder teams.

GENERATIVE AI PROJECTS

Leasel — Enterprise Generative AI Platform (Multi-Agent, RAG, HITL) Generative AI RAG Prompt Engineering

2026 · Python, LangGraph (8-agent), FastAPI, TF-IDF RAG, OCR, SQLite, REST APIs · github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built an enterprise-grade Generative AI platform using LLMs (OpenAI/Anthropic) and an 8-agent LangGraph orchestration pipeline — implements **intent classification** (Triage agent routes queries to the right specialist), **RAG** (TF-IDF retrieval with 100% citation correctness), **summarization** (structured root-cause summaries with ranked hypotheses), and conditional **HITL escalation** when AI confidence is too low.
- Engineered **prompt engineering** discipline throughout: chain-of-thought reasoning, few-shot examples, role specification, structured XML outputs, and output validation gates — 0% unsupported-claim rate, 0 safety boundary violations on full validation set including adversarial inputs.
- Achieved **93.6% precision and 87.9% F1** across 10 compliance rules on 1,240 lease PDFs; **CI/CD-integrated LLM evaluation** pipeline (LLM-as-judge, multi-trial sampling, regression gate) ensures regressions are blocked before merge; deployed via REST APIs with data privacy and governance enforced at every layer.

AI on FHIR — NLP-Driven Enterprise Data Query Platform LLM Integration REST API Automation

2024 · Flask, LLMs, FHIR REST APIs, React, Docker, spaCy, BERT, Embeddings · github.com/venkatalakshmiparimala/AI_on_FHIR

- Reduced manual data retrieval time ~60% by building a Generative AI automation platform integrating LLMs + BERT to translate natural language queries (EN/ES) into structured FHIR REST API calls over enterprise medical records — HIPAA-compliant RBAC enforced; deployed on Docker with full API integration into enterprise data sources.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering — JNTUA, India — 2013–2017

TECHNICAL SKILLS

Generative AI & LLMs: Generative AI, Large Language Models (LLMs), Prompt Engineering, Few-Shot Prompting, Zero-Shot Prompting, Chain-of-Thought Prompting, RAG, Retrieval-Augmented Generation, Intent Classification, Summarization, Embeddings, Semantic Search, Vector Database, ChromaDB, NLP, spaCy, BERT, Hugging Face, LangGraph, LangChain, Multi-Agent Orchestration, HITL, Output Validation, Hallucination Detection, OpenAI API, Anthropic API

Development & APIs: Python, REST APIs, FastAPI, Flask, React, TypeScript, JavaScript, SQL, PostgreSQL, MySQL, Git, GitHub Actions, CI/CD, Docker, Microservices, Automation, Bash scripting, OCR, Document Intelligence

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), Data Pipelines, ETL, RabbitMQ, Linux, Agile/Scrum

Governance & Security: Data Governance, Data Privacy, PHI Compliance, PII Handling, HIPAA, Access Controls, Audit Logging, Encryption (Libsodium), Regulated Environment AI